Advanced Calculus

Problem Set 4

Parameters, improper integrals, and logarithmic moments

Instructions. Show your work and justify all exchanges of limits and integrals.

1. Convergence of a parameter integral. (8 points)

Determine the values of the parameter for which the integral converges.

$$\int_0^1 x^{a}\,dx$$

2. Differentiation under the integral sign. (10 points)

State sufficient conditions for differentiating with respect to a parameter. Apply your result to the integral below and verify the answer directly.

$$F(t) = \int_0^1 e^{tx}\,dx$$

3. A limiting argument. (10 points)

Use a suitable convergence theorem to evaluate the following limit.

$$\lim_{m\to\infty} \int_0^1 \frac{mx}{1+mx}\,dx$$

4. A parameter-dependent rational integral. (10 points)

Evaluate the integral for a positive parameter, and explain why your expression may be differentiated with respect to that parameter.

$$F(a) = \int_0^1 \frac{1}{1+ax}\,dx, \qquad a > 0$$

5. Logarithmic moments. (12 points)

(a) For a real parameter n greater than -1, evaluate the integral

$$I_n = \int_0^1 x^n \ln x\,dx, \qquad n > -1$$

(b) Use your expression to evaluate the following two cases:

$$n = 2 \qquad\qquad n = 5$$

(c) Explain the sign of the integral without evaluating it.

6. A second logarithmic moment. (10 points)

Extend your argument from Problem 5 to determine a closed form for

$$J_n = \int_0^1 x^n (\ln x)^2\,dx, \qquad n > -1.$$

State clearly which hypotheses justify each step of your argument.